

# Supplemental Methods

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# 1. Genotype Parsing and Ploidy-Aware Haplotype Construction

## 1.1. Ploidy detection and genotype parsing

EasyHap infers each sample's operational ploidy from the maximum number of allele entries in its VCF GT fields:

$$\left[ p_s = \max_{j=1,\dots,m} |GT_{s,j}|, \right]$$

where  $(|GT_{s,j}|)$  is the number of allele at variant (j). Thus, samples with different ploidy levels can be analyzed in the same VCF. Genotype indices follow the VCF convention: 0 denotes REF, positive integers denote ALT alleles, and  $\cdot$  denotes a missing allele. Genotypes with fewer entries than  $(p_s)$  are padded with missing alleles in copy-resolved mode.

Phasing is determined from the genotype separator:  $|$  indicates phased data and  $/$  indicates unphased data. EasyHap uses the GT field for genotype or haplotype reconstruction and does not modify calls using genotype likelihoods, read depth, allele depth, or genotype quality.

EasyHap provides two analysis modes. The genotype mode performs phase-independent multilocus genotype analysis, whereas the copy mode reconstructs chromosome-copy haplotypes from phased genotypes. The legacy options `inbred` and `hybrid` are retained as aliases for `genotype` and `copy`, respectively.

## 1.2. Handling mixed ploidy, heterozygous genotypes, and missing data

Samples with different ploidy levels are processed independently. EasyHap neither converts samples to a common ploidy nor duplicates or removes chromosome copies. In copy mode, a sample with ploidy  $(p_s)$  contributes  $(p_s)$  chromosome-copy haplotypes; in genotype mode, each sample contributes one phase-independent multilocus genotype profile.

| Analysis mode | Input requirement                     | Output                                        | Heterozygote handling                     |
|---------------|---------------------------------------|-----------------------------------------------|-------------------------------------------|
| Genotype      | Phased or unphased genotypes          | One multilocus genotype profile per sample    | Allele order and phase are ignored        |
| Copy          | Heterozygous genotypes must be phased | $(p_s)$ chromosome-copy haplotypes per sample | Alleles are assigned to chromosome copies |

In genotype mode, allele states are converted to EasyHap tokens, missing entries are excluded, and the remaining distinct alleles are sorted before encoding. Thus, 0/1 and 1/0 are equivalent. Under the default `--hetero-policy slash`, an A/G heterozygote is

encoded as A/G. This representation records allele identity but not dosage; for example, 0/0/1 and 0/1/1 are equivalent if they contain the same distinct alleles.

Alternative heterozygote policies are available:

| Option  | Representation                                                               |
|---------|------------------------------------------------------------------------------|
| slash   | Distinct alleles joined by /                                                 |
| iupac   | IUPAC code for compatible nucleotide combinations; otherwise slash-separated |
| missing | Heterozygous sites converted to missing                                      |

A completely unresolved genotype is represented as N. In genotype mode, partially missing genotypes are summarized using their called alleles. In copy mode, missingness is retained separately for each chromosome copy. Copy mode requires phased heterozygous genotypes. EasyHap stops the analysis if it detects an unphased genotype containing multiple distinct nonmissing alleles. Homozygous or allele-uniform genotypes do not require phasing because copy assignment is unambiguous.

Ordinary missing data and structural absence are represented separately: N denotes an unresolved genotype, whereas ABS denotes sequence absence supported by an explicit deletion or absence event, as described in Section 2.4.

### 1.3. Copy-resolved haplotype construction from phased genotypes

Copy-resolved haplotypes are reconstructed from the ordered allele positions in phased VCF genotypes. For a sample with inferred ploidy ( $p_s$ ), EasyHap generates ( $p_s$ ) chromosome-copy sequences and assigns the ( $k$ )-th phased allele at each variant to copy ( $k$ ). Genotypes with fewer than ( $p_s$ ) alleles are padded with missing states. For sample ( $s$ ), copy ( $k$ ), and ( $m$ ) retained variants, the resulting haplotype can therefore be represented as

$$[H_{s,k} = (T_{1,a_{s,1,k}}, T_{2,a_{s,2,k}}, \dots, T_{m,a_{s,m,k}})]$$

where ( $a_{s,j,k}$ ) denotes the VCF allele index assigned to chromosome copy ( $k$ ) at variant ( $j$ ), and ( $T_{j,a}$ ) is the EasyHap token representing that allele. Missing alleles are represented by N; sequence absence caused by a deletion is encoded as ABS. For example, a phased diploid genotype series,

| Variant                | Phased genotype | Copy 1 | Copy 2 |
|------------------------|-----------------|--------|--------|
| 1                      | 0 1             | 0      | 1      |
| 2                      | 1 0             | 1      | 0      |
| 3                      | 0 0             | 0      | 0      |
| Copy-resolved sequence |                 | 0-1-0  | 1-0-0  |

The same procedure applies to triploid, tetraploid, and higher-ploidy samples, generating three, four, or more copy sequences. Identical copy-level sequences receive the same EasyHap haplotype identifier, and haplotype frequencies are calculated from chromosome copies. EasyHap does not perform phasing, phase correction, phase switching, or imputation, and it does not use PS fields to reconcile phase blocks. Copy-resolved analysis therefore requires consistent phased allele ordering across the analyzed interval.

## 2. Variant Representation and Integration

### 2.1. Encoding and normalization of SNPs, InDels, PAVs, and SVs

EasyHap converts VCF alleles into standardized marker states before haplotype construction. Canonical VCF normalization, including left alignment, trimming, and multiallelic decomposition, is not performed and should be completed upstream when required.

| Variant type | Representation in EasyHap                                                      | Main processing rule                                                                                      |
|--------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| SNP          | REF and ALT nucleotide states, with separate states for each ALT allele        | Multiallelic SNPs are retained as distinct allele states                                                  |
| InDel        | Token containing the allele index, length change, and ALT sequence information | Insertions and deletions are encoded as INS+length and DEL-length states                                  |
| PAV          | Genotype-bearing VCF allele, such as *, <DEL>, <CN0>, or <ABS>                 | * is encoded as ABS; downstream structural absence is inferred only when the deletion interval is defined |
| SV           | Symbolic or sequence-resolved VCF allele                                       | Symbolic labels and ALT indices are retained; SVLEN is included when available                            |

PAVs and SVs must be explicitly represented in the genotype-bearing VCF. EasyHap does not infer them from read depth, breakpoint evidence, assemblies, or separate presence/absence matrices. Deletion-like alleles are identified from \*, deletion/absence symbolic labels, sequence-resolved ALT alleles shorter than REF, or SVTYPE=DEL. The deletion endpoint is obtained from END when available; otherwise, the REF span is used.

## 2.2. Handling multiallelic, overlapping, and nested variants

Multiallelic records are retained as single VCF markers, and each ALT allele preserves its original allele index. For example, REF=A, ALT=C,G,T is represented as four distinct states: 0=A, 1=C, 2=G, and 3=T. Polyploid genotypes are processed using these allele indices without reducing the site to a REF-versus-ALT comparison. Overlapping and nested variants are not merged into synthetic compound alleles. Each VCF record remains an independent marker, and overlap information is reported in VariantOverlap.tsv.

| Situation                                   | EasyHap handling                                                                  |
|---------------------------------------------|-----------------------------------------------------------------------------------|
| Multiallelic record                         | Retained as one marker; all allele indices are preserved                          |
| Records with the same start position        | Retained independently                                                            |
| Overlapping or nested variants              | Retained independently and reported                                               |
| Marker inside an explicit deletion interval | Converted to structural absence (ABS) for chromosome copies carrying the deletion |
| Complex SVs or shared breakpoints           | Not reconciled into genome graphs; records remain separate                        |

For recognized deletions, the genomic interval is determined from the REF span and, when available, the deletion endpoint. The deletion itself remains an independent marker, while internal markers may be assigned ABS. This prevents the same large deletion from being counted repeatedly through all downstream variants. Because EasyHap does not resolve complex overlapping SVs into alternative genome graphs, highly complex loci should be represented using a consistently curated VCF and interpreted at the individual-record level.

## 2.3. Joint analysis of multiple variant types

After tokenization, all retained variants are represented in a common ordered marker space regardless of whether the original record is a SNP, InDel, PAV, or SV. A haplotype or phase-independent genotype profile is therefore a vector containing one state per retained VCF record. No separate SNP, InDel, and SV haplotypes are constructed and merged afterwards.

For example, an interval may contain the following ordered markers:

SNP – InDel – multiallelic SNP – deletion/PAV – SNP – SNP

Each sample or chromosome copy receives one token at each position. The complete token vector is then used to identify unique haplotypes and to calculate sequence distances. This permits SNPs, InDels, and appropriately represented PAV/SV alleles to contribute jointly to regional haplotype definition.

Each VCF record contributes one marker to the comparison; EasyHap does not weight a variant according to its physical length or variant class. Thus, a 1-bp SNP and a 500-bp deletion-defining record each contribute one directly comparable event. For a large deletion, however, markers lying within the deleted interval can be marked ABS and excluded from pairwise distance calculations, preventing the deletion from being counted repeatedly through multiple downstream absent SNPs.

## 2.4. Distinguishing structural absence from ordinary missing genotypes

EasyHap distinguishes two internal states:

| State | Meaning                        | Interpretation                                          |
|-------|--------------------------------|---------------------------------------------------------|
| N     | Missing or unresolved genotype | Sequence is presumed present, but the allele is unknown |
| ABS   | Structural absence             | Sequence is absent because of a deletion/PAV            |

ABS is assigned only when an explicit deletion/PAV record is available and its interval can be determined. Missing downstream SNP or InDel genotypes alone are not sufficient evidence of structural absence.

In copy mode, deletion status is propagated to downstream markers within the deletion interval:

| Deletion genotype                        | Propagation rule                                   |
|------------------------------------------|----------------------------------------------------|
| All chromosome copies carry the deletion | Mark affected copies as ABS                        |
| Heterozygous phased deletion             | Mark only the phased deletion-carrying copy as ABS |
| Heterozygous unphased deletion           | Do not assign ABS to a specific copy               |

For example, a phased diploid genotype 0|1 at a deletion with END=500, where ALT1 is <DEL>, identifies copy 2 as deletion-carrying. Variants at positions 200–400 are therefore assigned ABS on copy 2, while copy 1 retains its observed genotype or N.

In genotype mode, homolog-specific assignment is unavailable for an unphased heterozygous deletion such as 0/1. The deletion allele is retained at the deletion marker, but downstream ABS states are propagated only when all non-missing alleles indicate deletion. This avoids treating phase uncertainty as structural evidence.

During distance calculation, both N and downstream ABS positions are excluded from pairwise comparisons. The deletion-defining marker remains comparable, so a

large deletion is counted as one structural event rather than as multiple differences across the deleted interval. Structural absence cannot be inferred when:

- only downstream genotypes are missing;
- a symbolic deletion lacks sufficient interval information;
- the deletion-defining record is outside the analyzed region and is not loaded.

Such cases require upstream SV/PAV harmonization or inclusion of the causal structural-variant record in the analysis interval.

### 3. Haplotype Clustering and Analysis

#### 3.1. Sequence representation and distance calculation

Clustering is performed on the original EasyHap marker-token vectors rather than on reconstructed nucleotide sequences or on the single-character FASTA/PHYLIP/NEXUS export representation. For each haplotype, the sequence used for clustering is

$$[H = (h_1, h_2, \dots, h_m)]$$

where ( $h_j$ ) is the allele/genotype token at variant ( $j$ ). For two sequences ( $H_a$ ) and ( $H_b$ ), EasyHap calculates a normalized Hamming distance over pairwise comparable marker positions:

$$\left[ d(H_a, H_b) = \frac{\sum_{j \in C} I(h_{a,j} \neq h_{b,j})}{|C|} \right]$$

where ( $C$ ) is the set of marker positions for which neither sequence contains an ordinary missing state ( $N$ ) nor a structural-absence state ( $ABS$ ). ( $I(\cdot)$ ) is an indicator function equal to 1 when the two states differ and 0 when they are identical.

The distance therefore ranges from 0 to 1. A value of 0 indicates identical states at every comparable marker, whereas a value of 1 indicates that every comparable marker differs. If two sequences have no comparable markers, EasyHap assigns a distance of 1 rather than treating them as similar.

Comparison is performed on complete marker states. In genotype mode, for example, A/G is different from A, but A/G and a genotype originally written as G/A are identical because allele order is normalized before sequence construction. As noted above, genotype mode compares allele sets rather than dosage; therefore, polyploid states containing the same set of alleles but different allele copy numbers are not distinguished by this representation.

The single-character alignment files generated by EasyHap are downstream export formats. At each marker, nucleotide states are preserved where possible and other allele tokens are assigned site-specific character codes, with the original-to-encoded relationship recorded separately. These encoded alignment characters are not used to calculate the EasyHap clustering distance.

### 3.2. Clustering algorithm, linkage method, and normalization

EasyHap does not use conventional hierarchical clustering with Ward, average, complete, or UPGMA linkage. Instead, it applies a distance-threshold graph procedure.

First, normalized Hamming distances are calculated for every pair of unique haplotypes. Two haplotypes are connected by an undirected edge when

$$[d(H_i, H_j) \leq \tau]$$

where ( $\tau$ ) is the user-defined --cluster-threshold. The default value is 0.15. Second, EasyHap identifies the connected components of this graph using a union-find procedure. Every haplotype belonging to the same connected component receives the same cluster identifier (HapC1, HapC2, etc.).

This procedure is operationally analogous to thresholded single-linkage connectivity: if haplotype A is sufficiently similar to B and B is sufficiently similar to C, all three can belong to the same component even when the direct A–C distance exceeds the threshold. EasyHap does not construct a dendrogram and does not optimize a hierarchical linkage objective. Normalization occurs directly in the distance definition. The number of differences is divided by the number of markers actually comparable for that particular haplotype pair. No additional scaling by chromosome length, allele frequency, variant class, or physical SV size is applied.

### 3.3. Treatment of missing dat

Missing information is handled by pairwise deletion during clustering. A marker is excluded from a pairwise comparison if either haplotype contains N or ABS at that position. The denominator of the normalized Hamming distance is therefore the number of markers comparable for that specific pair rather than the total number of variants in the interval.

No imputation of missing alleles is performed, and missing genotypes are not automatically treated as reference alleles. Similarly, structural absence is not treated as an ordinary alternative nucleotide at every marker inside a deletion. The deletion-defining variant can distinguish the two haplotypes, whereas the structurally unavailable downstream markers are excluded.

This strategy prevents missingness and large deletions from creating artificial strings of sequence differences, but it also means that distance estimates based on only a small number of mutually observed markers can be less informative. When no comparable marker remains, the distance is explicitly set to 1. Users working with datasets containing extensive missingness should therefore inspect genotype completeness and consider whether the selected interval contains sufficient jointly observed markers.

### 3.4. Applicability to phased and unphased genotype data

EasyHap explicitly separates analyses that require haplotype phase from analyses that do not.

The genotype mode accepts both phased and unphased VCF genotypes. Phase separators are ignored when defining the site state, so 0|1, 0/1, and 1/0 generate the same phase-independent heterozygous state under the default encoding. The resulting EasyHap “Hap” classes in this mode should therefore be interpreted as unique multilocus genotype or allelic-configuration classes rather than homolog-specific molecular haplotypes. This mode is useful when copy-resolved phase is unavailable or unnecessary, including many analyses dominated by homozygous material, PAV presence/absence states, or unphased multilocus genotype combinations. Haplotype-style heatmaps and population summaries can still be produced, but the visualization represents phase-independent multilocus configurations.

The copy mode is intended for true chromosome-copy-resolved reconstruction. Every heterozygous genotype in the retained interval must be phased. EasyHap does not run a phasing algorithm internally. If an unphased heterozygous genotype is detected, users must either analyze the data in genotype mode or phase the VCF using a dedicated external phasing method before rerunning copy mode. The requirement is deliberately genotype-specific rather than file-wide: unphased homozygous genotypes are acceptable because their chromosome-copy assignment is unambiguous. However, EasyHap currently uses only the phased allele order and does not model phase-set boundaries. Therefore, multiple independently phased blocks should not be assumed to form a chromosome-wide haplotype unless their phase orientation has been made consistent upstream.

For unphased heterozygous PAVs/SVs, genotype mode retains the deletion/PAV allele as part of the multilocus genotype profile but does not assign downstream structural absence to a particular homolog. This is a deliberate limitation intended to prevent unsupported copy assignment.

## 4. Input Formats, Supported Scenarios, and Limitations

### 4.1. VCF/BCF and PLINK input formats

The primary EasyHap input is a genotype-containing VCF, compressed VCF, or BCF file, including standard genotype fields and, when available, SV annotations such as SVTYPE, END, and SVLEN. Indexed VCF/BCF files can be accessed regionally using `cyvcf2` or `pysam`, whereas unindexed VCF files are processed sequentially.

EasyHap v1.3.0 also supports PLINK 1 binary datasets (.bed, .bim, and .fam) through `--bfile`. PLINK input is limited to unphased, biallelic genotype analysis and is therefore restricted to phase-independent genotype mode. For internal decoding, BIM allele 1 (A1) is treated as ALT and allele 2 (A2) as REF, while the original allele identities are retained in the metadata.

Because PLINK does not preserve multiallelic variants, homolog-specific phase, arbitrary polyploid genotypes, or detailed SV annotations, VCF/BCF remains the preferred format for copy-resolved haplotype reconstruction and analyses involving multiallelic variants, PAVs, or SVs.

## **4.2. Supported ploidy levels and genotype configurations**

The VCF parser supports haploid, diploid, triploid, tetraploid, and higher-order genotypes by determining ploidy from the number of alleles in each sample's GT field. Samples within the same VCF may therefore have different ploidies. Copy mode reconstructs the corresponding number of chromosome-copy sequences per sample, whereas genotype mode generates one phase-independent multilocus profile per accession.

EasyHap does not distinguish auto- from allopolyploidy, infer subgenome assignments, or estimate allele dosage from sequencing reads; this information must be encoded in GT. Within a sample, variable genotype lengths are handled using the maximum observed GT length in the analyzed interval, with shorter genotypes padded as missing in copy mode. This is a technical strategy for incomplete input, not a model of mosaic ploidy or aneuploidy.

Genotype mode ignores phase and allele dosage after converting heterozygous calls to unique allele sets. Copy mode preserves chromosome-copy information but requires reliably phased heterozygous genotypes with consistent copy orientation across the analyzed interval.

## **4.3. Information requirements for SNP/InDel, PAV, and SV analyses**

For SNPs and sequence-resolved InDels, EasyHap requires standard VCF positional information, REF/ALT alleles, and sample GT fields, and supports both biallelic and multiallelic records.

PAVs/SVs must be explicitly represented in the genotype data. Sequence-resolved deletions are identified from REF/ALT length differences, while symbolic alleles such as <DEL>, <CN0>, <ABS>, and \* are treated as deletion or absence states. A reliable END field is recommended for large deletions; without it, the extent of symbolic SVs may not be determined accurately.

Other symbolic SVs are retained as marker states, but EasyHap does not reconstruct complex rearrangements, translocations, inversions, copy-number dosage, or nested SV architectures. Structural absence is inferred only from an explicit SV genotype and interval information; missing downstream SNPs alone are not sufficient. PLINK data without SV interval metadata cannot support this inference.

#### 4.4. Scope and limitations of EasyHap

EasyHap is a regional genotype/haplotype analysis framework, not a variant caller, genotype imputation tool, phasing program, or general SV reconstruction engine. Its main limitations are summarized below.

| Aspect                | Limitation                                                                                                                                                                                                                                                                        |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ploidy                | Ploidy is inferred from the GT structure provided in the input. EasyHap does not infer biological ploidy from raw sequencing data or validate cytogenetic ploidy.                                                                                                                 |
| Phasing               | Copy-resolved haplotype analysis requires consistently phased heterozygous genotypes. EasyHap does not perform phasing or resolve phase-set boundaries.                                                                                                                           |
| Genotype mode         | Heterozygous allele identities are retained, but allele dosage is not. Polyploid genotypes with the same allele set but different dosages may therefore be treated as equivalent.                                                                                                 |
| Variant normalization | EasyHap standardizes allele states but does not perform canonical VCF left normalization, multiallelic decomposition, or duplicate-event reconciliation. Input VCF harmonization is recommended.                                                                                  |
| PAV/SV detection      | Structural absence (ABS) requires an explicit deletion/PAV record with a known or derivable interval. Missing downstream genotypes alone are not interpreted as structural absence. Complex SVs and symbolic deletions lacking endpoint information may not be resolved reliably. |
| Missing data          | Missing and structurally absent markers are excluded from pairwise distance calculations. No imputation or explicit missing-data penalty is applied, and distances based on few comparable markers may be less reliable.                                                          |
| PLINK input           | PLINK BED/BIM/FAM is supported mainly for conventional biallelic, unphased genotype analysis. VCF/BCF is required for multiallelic variants, phased copy reconstruction, polyploid allele tuples, and structurally informative PAV/SV metadata.                                   |
| Clustering            | Clustering uses normalized Hamming-distance thresholding and connected components rather than hierarchical clustering. Results may therefore depend on the selected distance threshold and indirect graph connections.                                                            |

Accordingly, EasyHap supports two distinct analytical applications: (i) phase-independent regional multilocus genotype analysis and (ii) copy-resolved haplotype reconstruction from consistently phased genotypes. These applications should not be interpreted interchangeably, particularly when allele dosage, phase, or complex structural variation is biologically important.

## 5. Benchmarking and Comparison with Existing Tools

### 5.1. Validation of computational correctness and robustness

EasyHap does not perform variant calling, genotype imputation, or statistical phasing; therefore, haplotype accuracy depends on the quality of the input genotypes and, for copy-resolved analysis, upstream phasing. We evaluated computational correctness using synthetic datasets with predefined haplotypes covering different ploidy levels, heterozygous and multiallelic genotypes, SNPs, InDels, missing data, and PAV/SV-like deletions.

As summarized in the table below, EasyHap correctly reproduced the predefined results across all supported scenarios and appropriately detected unsupported cases, such as unphased heterozygous genotypes in copy-resolved mode. These results demonstrate accurate and reproducible genotype-to-haplotype reconstruction when valid genotype and phase information are provided.

| Validation scenario               | Expected                     | EasyHap output | Reconstruction concordance | Result |
|-----------------------------------|------------------------------|----------------|----------------------------|--------|
| Haploid                           | 6 haplotypes                 | 6 haplotypes   | 100%                       | PASS   |
| Diploid phased                    | 6 haplotypes                 | 6 haplotypes   | 100%                       | PASS   |
| Triploid phased                   | 6 haplotypes                 | 6 haplotypes   | 100%                       | PASS   |
| Tetraploid phased                 | 6 haplotypes                 | 6 haplotypes   | 100%                       | PASS   |
| Multiallelic variants             | predefined alleles           | identical      | 100%                       | PASS   |
| SNP + InDel                       | predefined haplotypes        | identical      | 100%                       | PASS   |
| SNP + InDel + PAV                 | predefined haplotypes        | identical      | 100%                       | PASS   |
| Missing genotype                  | predefined missing positions | identical      | 100%                       | PASS   |
| Structural absence                | expected ABS propagation     | identical      | 100%                       | PASS   |
| Unphased heterozygote + copy mode | error expected               | error reported | /                          | PASS   |

\*Reconstruction concordance =  $\frac{\text{number of correctly reconstructed haplotype states}}{\text{total expected haplotype states}} \times 100\%$

## 5.2. Runtime and memory benchmark

For small datasets, runtime is strongly affected by platform-specific VCF access overhead, whereas for larger datasets the downstream haplotype analysis becomes the dominant computational cost, reducing the relative runtime difference between the Linux/cyvcf2 and Windows/bcftools implementations.

Computational performance of EasyHap v1.3 on the Linux platform

| Samples | Retained variants | OS/backend            | Runtime (s) | Peak RAM (GB) |
|---------|-------------------|-----------------------|-------------|---------------|
| 100     | 50                | Ubuntu 22.04 / cyvcf2 | 0.33        | 0.066         |
| 500     | 50                | Ubuntu 22.04 / cyvcf2 | 0.43        | 0.07          |
| 1,000   | 50                | Ubuntu 22.04 / cyvcf2 | 0.62        | 0.074         |
| 2,000   | 50                | Ubuntu 22.04 / cyvcf2 | 1.07        | 0.07          |
| 3,000   | 50                | Ubuntu 22.04 / cyvcf2 | 1.59        | 0.094         |
| 4,000   | 100               | Ubuntu 22.04 / cyvcf2 | 7.55        | 0.121         |
| 4,000   | 200               | Ubuntu 22.04 / cyvcf2 | 33.07       | 0.176         |
| 4,000   | 500               | Ubuntu 22.04 / cyvcf2 | 710.6       | 0.344         |

**\*Note:** Performance was evaluated using the basic EasyHap analysis workflow without plotting. Linux tests were performed on Ubuntu 22.04 with dual AMD EPYC 9754 CPUs (256 cores/512 threads) and 1.5 TiB RAM.

Computational performance of EasyHap v1.3 on the Windows platform

| Samples | Retained variants | OS/backend            | Runtime (s) | Peak RAM (GB) |
|---------|-------------------|-----------------------|-------------|---------------|
| 100     | 50                | Windows 10 / bcftools | 1.93        | 0.226         |
| 500     | 50                | Windows 10 / bcftools | 8.32        | 0.232         |
| 1,000   | 50                | Windows 10 / bcftools | 15.86       | 0.239         |
| 2,000   | 50                | Windows 10 / bcftools | 30.81       | 0.230         |
| 3,000   | 50                | Windows 10 / bcftools | 50.22       | 0.261         |
| 4,000   | 100               | Windows 10 / bcftools | 72.85       | 0.320         |
| 4,000   | 200               | Windows 10 / bcftools | 89.36       | 0.399         |
| 4,000   | 500               | Windows 10 / bcftools | 675.16      | 0.588         |

**\*Note:** Performance was evaluated using the basic EasyHap analysis workflow without plotting. Windows tests were performed on 64-bit Windows 10 Pro with an Intel Core i5-12600K CPU (10 cores/16 threads) and 31.78 GB RAM.

Runtime and peak memory usage comparison among EasyHap, CandiHap, HAPPE, geneHapR, and crosshap using 4,000 samples and 100 retained variants without plotting.

| Metric                   | EasyHap<br>v1.3.0        | CandiHap<br>v1.3.2         | geneHapR<br>v1.2.6  | crosshap<br>v1.4.0                 | HAPPE<br>v0.1.5            |
|--------------------------|--------------------------|----------------------------|---------------------|------------------------------------|----------------------------|
| Environment<br>/ backend | Ubuntu 22.04<br>/ cyvcf2 | Ubuntu 22.04<br>/ Perl + R | Ubuntu 22.04<br>/ R | Ubuntu 22.04 /<br>R + PLINK<br>1.9 | Ubuntu 22.04<br>/ bcftools |
| Runtime (s)              | 7.55                     | 3.55                       | 8.89                | 3.46                               | 113.23                     |
| Peak RAM<br>(GB)         | 0.121                    | 0.119                      | 0.850               | 0.215                              | 0.203                      |

**\*Note:** Runtime was measured for the non-plotting analytical workflow. all tools were benchmarked using the same pre-extracted regional VCF containing 4,000 samples and 100 retained variants. For crosshap, the reported runtime includes the required PLINK-based LD-matrix calculation and subsequent haplotyping analysis. For CandiHap, the reported runtime includes VCF-to-HMP conversion and the CandiHap haplotype-analysis workflow, while ANNOVAR annotation was excluded. Common regional VCF extraction was excluded from the crosshap and CandiHap timings. For HAPPE, the reported runtime (113.23 s) represents the main computational analysis only. HAPPE additionally writes haplotype information for individual samples to Excel files; for this 4,000-sample dataset, this output step required several additional hours and was therefore not included in the reported computational runtime.

### 5.3. Feature comparison with CandiHap, HAPPE, geneHapR, and crosshap

| Feature                            | EasyHap                                                          | CandiHap                                                       | HAPPE                                             | geneHapR                                             | crosshap                                 |
|------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------|------------------------------------------|
| Primary input                      | VCF/VCF.GZ/BCF;<br>PLINK<br>BED/BIM/FAM                          | VCF/HapMap;<br>Sanger<br>data                                  | bgzip-<br>compressed<br>VCF                       | VCF, FASTA,<br>PLINK PED/MAP,<br>HapMap, tables      | VCF +<br>pairwise LD<br>matrix           |
| Ploidy-aware<br>analysis           | ✓ Haploid to<br>polyploid; sample-<br>specific ploidy<br>from GT | — Not<br>explicitly<br>ploidy-aware                            | — Not<br>explicitly<br>ploidy-aware               | — Not explicitly<br>ploidy-aware                     | — Diploid<br>genotype-state<br>framework |
| Mixed ploidy<br>within one dataset | ✓                                                                | —                                                              | —                                                 | —                                                    | —                                        |
| Heterozygous<br>genotypes          | ✓ Explicit<br>genotype- and<br>copy-mode<br>handling             | △<br>Applicable to<br>heterozygotes<br>for gene-<br>haplotype. | △ Encoded as<br>a single<br>heterozygous<br>state | △ Can retain or<br>remove<br>heterozygous<br>samples | ✓<br>Heterozygous<br>state included      |

|                                                |                                                                    |                                     |                                                              |                                                     |                                               |
|------------------------------------------------|--------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------|
| <b>Phased copy-resolved haplotypes</b>         | ✓                                                                  | —                                   | —                                                            | —                                                   | —                                             |
| <b>Unphased genotype analysis</b>              | ✓ Genotype mode                                                    | ✓                                   | ✓                                                            | ✓                                                   | ✓                                             |
| <b>SNPs</b>                                    | ✓                                                                  | ✓                                   | ✓                                                            | ✓                                                   | ✓                                             |
| <b>InDels</b>                                  | ✓                                                                  | ✓                                   | △ Not a primary haplotyping focus                            | ✓                                                   | — Primarily SNP-based                         |
| <b>Multiallelic variants</b>                   | ✓ Preserves individual ALT alleles                                 | — biallelic variants                | — Not explicitly supported                                   | — Not explicitly emphasized                         | — Biallelic SNP-state framework               |
| <b>PAV/SV integration</b>                      | ✓ Genotype-bearing PAV/SV alleles                                  | —                                   | △ Read-depth information can assist CNV/InDel interpretation | —                                                   | —                                             |
| <b>Structural absence vs. missing genotype</b> | ✓ ABS and N treated separately                                     | —                                   | —                                                            | —                                                   | —                                             |
| <b>Haplotype/similarity clustering</b>         | ✓ Normalized Hamming distance + threshold clustering               | △ Identical gene-haplotype grouping | ✓ Euclidean distance; Ward hierarchical clustering           | △ Haplotype/network analysis                        | △ DBSCAN clustering of LD-based Marker Groups |
| <b>Population/group comparison</b>             | ✓ Group frequency, Hd, private haplotypes                          | △ Basic haplotype/sample statistics | △ Sample metadata visualization                              | ✓ Frequency and geographical distribution           | ✓ Metadata/group composition                  |
| <b>Trait association/comparison</b>            | ✓ Haplotype–trait comparison and significance tests                | ✓                                   | △ Trait/GWAS information can be displayed                    | ✓                                                   | ✓                                             |
| <b>LD analysis/visualization</b>               | ✓ Regional LD calculation and heatmap                              | ✓ LD-region analysis                | —                                                            | ✓ LD-block analysis                                 | ✓ LD-based grouping; LD matrix required       |
| <b>Integrated gene+haplotype visualization</b> | ✓ Gene structure, haplotype matrix, LD, population and trait plots | ✓                                   | △ Editable Excel haplotype                                   | △ Haplotype heatmap, LD, population and trait plots | △ Haplotype heatmap                           |
| <b>User interface</b>                          | Linux CLI + Windows GUI                                            | CLI + GUI                           | CLI; editable Excel output                                   | R package                                           | R package                                     |

|                                    |                                                                                 |                                           |                                                 |                                                              |                                                              |
|------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| <b>Main distinguishing feature</b> | <b>Ploidy-aware, mixed-ploidy and multi-variant regional haplotype analysis</b> | Candidate-gene/favorable-haplotype mining | Editable Excel genotype/haplotype visualization | Comprehensive R-based haplotype statistics and visualization | LD-based dimensional reduction and local trait visualization |
|------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|

Symbols: ✓, explicitly supported/core functionality; △, partially supported, indirect support, or not the principal purpose of the tool; —, not explicitly supported or described as a core capability in the corresponding publication/documentation.

Compared with existing local haplotype tools, EasyHap is distinguished primarily by its ploidy-aware design, explicit separation of phase-independent and copy-resolved analyses, support for multiallelic and heterogeneous variant types, and treatment of structural absence separately from ordinary missing genotypes.

## 6. Worked Examples and Case Studies

These deterministic synthetic datasets were redesigned for manuscript/reviewer validation. Each example has approximately 20 variant records and is built around six known major haplotypes. In the intended EasyHap mode, all six are observed in more than five accessions and also exceed 5% frequency. These example files could be retrieved from [https://github.com/Guangqi-He/EasyHap/blob/main/examples/reviewer\\_validation.zip](https://github.com/Guangqi-He/EasyHap/blob/main/examples/reviewer_validation.zip).

| Dataset                                  | Samples | Variants | Mode     | Expected result                                                  |
|------------------------------------------|---------|----------|----------|------------------------------------------------------------------|
| diploid_high_heterozygosity_unphased.vcf | 42      | 20       | genotype | 6 haplotypes; high heterozygosity; copy mode rejected            |
| diploid_phased_heterozygous.vcf          | 36      | 20       | copy     | 6 phased copy haplotypes; 12 accessions/haplotype                |
| autotetraploid.vcf                       | 36      | 20       | copy     | 6 phased copy haplotypes in four-copy genotypes                  |
| allotetraploid_subgenomes.vcf            | 36      | 20       | copy     | 10 A + 10 B sites; 6 abundant haplotypes/subgenome               |
| large_deletion.vcf                       | 36      | 20       | copy     | 6 copy haplotypes incl. 2 deletion haplotypes; ABS ≠ missing     |
| multiallelic_overlap.vcf                 | 36      | 20       | copy     | 6 copy haplotypes across multiallelic/overlapping/nested records |
| plink_toy.bed/.bim/.fam                  | 42      | 20       | genotype | 6 PLINK genotype haplotypes; 7 accessions/haplotype              |

## 6.1. Multiallelic and mixed-variant encoding examples

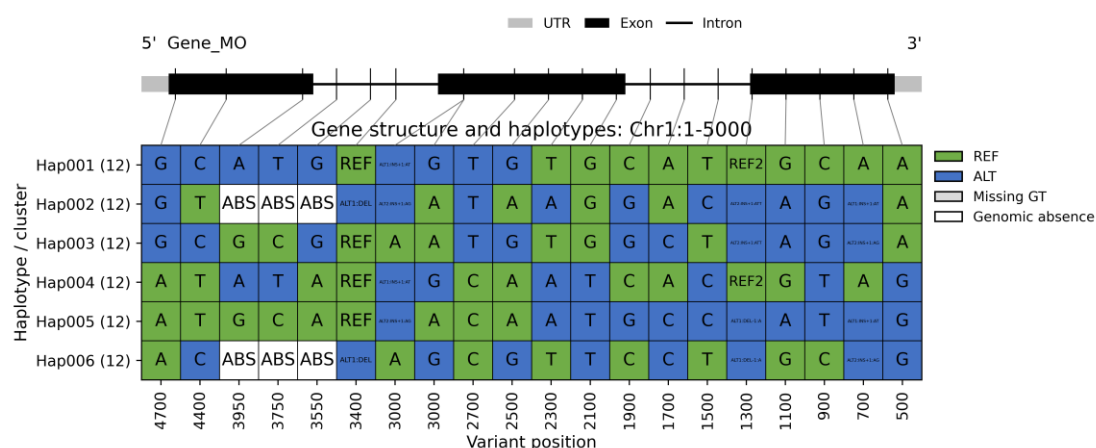
Input VCF: multiallelic\_overlap.vcf

```
##fileformat=VCFv4.2
##contig=<ID=Chr1>
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="SV type">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT MOC1_01
Chr1 500 mo01A G . PASS. GT 0|1
Chr1 700 sameLenINS A AT,AG . PASS. GT 0|1
Chr1 900 triallelicSNP C T,G . PASS. GT 0|1
Chr1 1100mo04G A . PASS. GT 0|1
Chr1 1300multiIndel AT A,ATT . PASS. GT 0|1
Chr1 1500mo06T C . PASS. GT 0|1
Chr1 1700mo07A C . PASS. GT 0|1
Chr1 1900mo08C G . PASS. GT 0|1
Chr1 2100mo09G T . PASS. GT 0|1
Chr1 2300mo10T A . PASS. GT 0|1
Chr1 2500mo11A G . PASS. GT 1|0
Chr1 2700mo12C T . PASS. GT 1|0
Chr1 3000sameStartSNP A G . PASS. GT 1|0
Chr1 3000sameStartINS A AT,AG . PASS. GT 1|2
Chr1 3400nestedDEL N <DEL> . PASS SVTYPE=DEL;END=4100 GT 0|0
Chr1 3550nestedSNP1 A G . PASS. GT 1|0
Chr1 3750nestedSNP2 C T . PASS. GT 1|0
Chr1 3950nestedSNP3 G A . PASS. GT 1|0
Chr1 4400post01 T C . PASS. GT 1|0
Chr1 4700post02 A G . PASS. GT 1|0
```

Command

```
easyhap analyze --vcf multiallelic_overlap.vcf --gff multiallelic_overlap.gff3 -
-region Chr1:1-5000 --mode copy --plot --plot-format png --outdir out_overlap
```

Results



The gene is located on the negative strand; therefore, EasyHap reverses the plotting orientation to display the gene from 5' to 3' from left to right. Consequently, variant coordinates are shown in descending genomic order.

## 6.2. Large-deletion example with loss of deletion-overlapping markers

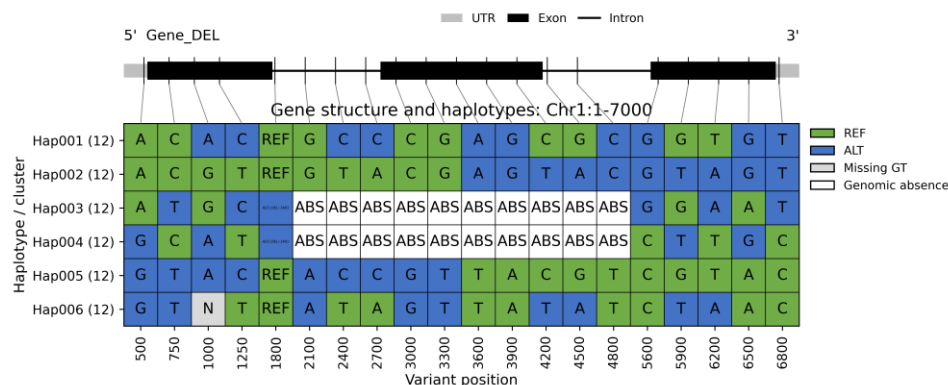
Input VCF: large\_deletion.vcf

```
##fileformat=VCFv4.2
##contig=<ID=Chr1>
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="SV type">
##INFO=<ID=SVLEN,Number=1,Type=Integer,Description="SV length">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT DLC1_01
Chr1 500 pre01 A G . PASS . GT 0|1
Chr1 750 pre02 C T . PASS . GT 0|1
Chr1 1000 pre03 G A . PASS . GT 0|1
Chr1 1250 pre04 T C . PASS . GT 0|1
Chr1 1800 DEL_LARGE_01 N <DEL> . PASS SVTYPE=DEL;END=5200;SVLEN=-3401 GT 0|0
Chr1 2100 inDEL01 G A . PASS . GT 0|1
Chr1 2400 inDEL02 T C . PASS . GT 0|1
Chr1 2700 inDEL03 A C . PASS . GT 0|1
Chr1 3000 inDEL04 C G . PASS . GT 0|1
Chr1 3300 inDEL05 G T . PASS . GT 0|1
Chr1 3600 inDEL06 T A . PASS . GT 1|0
Chr1 3900 inDEL07 A G . PASS . GT 1|0
Chr1 4200 inDEL08 C T . PASS . GT 1|0
Chr1 4500 inDEL09 G A . PASS . GT 1|0
Chr1 4800 inDEL10 T C . PASS . GT 1|0
Chr1 5600 post01 C G . PASS . GT 1|0
Chr1 5900 post02 G T . PASS . GT 1|0
Chr1 6200 post03 T A . PASS . GT 1|0
Chr1 6500 post04 A G . PASS . GT 1|0
Chr1 6800 post05 C T . PASS . GT 1|0
```

Command

```
easyhap analyze --vcf large_deletion.vcf --gff large_deletion.gff3 --region
Chr1:1-7000 --mode copy --plot --plot-format png --outdir out_large_del
```

Results



This example shows how EasyHap handles a heterozygous Chr1:1800–5200 deletion in copy mode: markers within the deleted interval are treated as structurally absent on the deletion-bearing haplotype, while flanking markers and markers on the intact haplotype remain informative, distinguishing genomic absence from ordinary missing genotypes.

### 6.3. Polyploid and complex-genotype case study

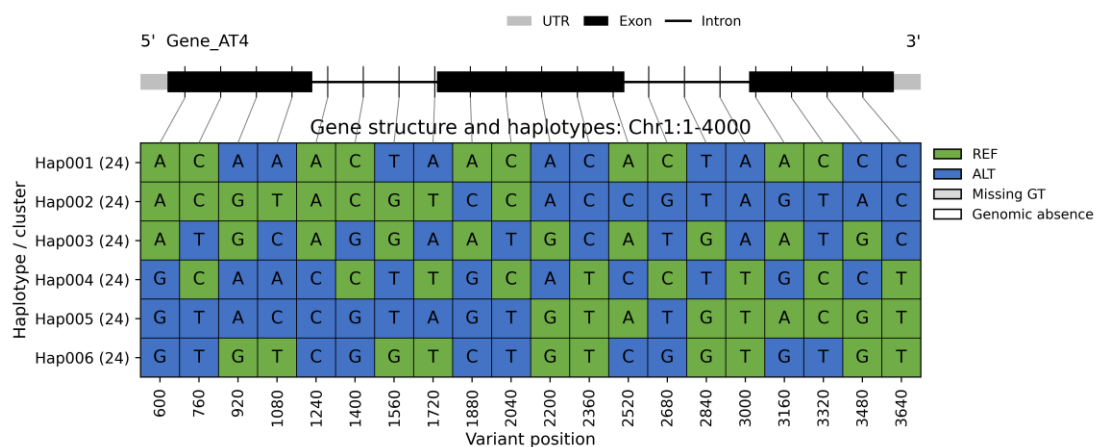
Input VCF: autotetraploid.vcf

```
##fileformat=VCFv4.2
##contig=<ID=Chr1>
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT ATC1_01
Chr1 600 at01A G . PASS . GT 0|1|0|1
Chr1 760 at02C T . PASS . GT 0|1|0|1
Chr1 920 at03G A . PASS . GT 0|1|1|0
Chr1 1080 at04T C,A . PASS . GT 0|1|2|0
Chr1 1240 at05A C . PASS . GT 0|1|0|1
Chr1 1400 at06C G . PASS . GT 0|1|0|1
Chr1 1560 at07G T . PASS . GT 0|1|1|0
Chr1 1720 at08T A . PASS . GT 0|1|1|0
Chr1 1880 at09A G,C . PASS . GT 2|1|0|2
Chr1 2040 at10C T . PASS . GT 0|1|0|1
Chr1 2200 at11G A . PASS . GT 1|0|1|0
Chr1 2360 at12T C . PASS . GT 1|0|1|0
Chr1 2520 at13A C . PASS . GT 1|0|0|1
Chr1 2680 at14C G,T . PASS . GT 1|2|0|1
Chr1 2840 at15G T . PASS . GT 1|0|1|0
Chr1 3000 at16T A . PASS . GT 1|0|1|0
Chr1 3160 at17A G . PASS . GT 1|0|0|1
Chr1 3320 at18C T . PASS . GT 1|0|0|1
Chr1 3480 at19G A,C . PASS . GT 1|0|2|0
Chr1 3640 at20T C . PASS . GT 1|0|1|0
```

Command

```
easyhap analyze --vcf autotetraploid.vcf --gff autotetraploid.gff3 --region
Chr1:1-4000 --mode copy --plot --plot-format png --outdir out_auto4x
```

Results



This example shows that EasyHap can reconstruct four copy-specific haplotypes from phased autotetraploid genotypes while retaining biallelic and multiallelic variation.

Allotetraploid: analyze the two subgenomes separately

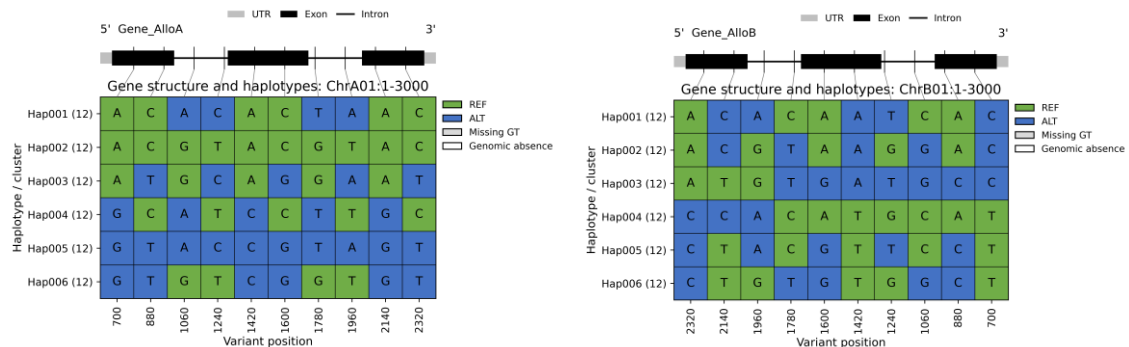
Input VCF: allotetraploid\_subgenomes.vcf

```
##fileformat=VCFv4.2
##contig=<ID=ChrA01>
##contig=<ID=ChrB01>
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT ALC1_01
ChrA01 700 A01 A G . PASS . GT 0|1
ChrA01 880 A02 C T . PASS . GT 0|1
ChrA01 1060 A03 G A . PASS . GT 0|1
ChrA01 1240 A04 T C . PASS . GT 0|1
ChrA01 1420 A05 A C . PASS . GT 0|1
ChrA01 1600 A06 C G . PASS . GT 0|1
ChrA01 1780 A07 G T . PASS . GT 0|1
ChrA01 1960 A08 T A . PASS . GT 0|1
ChrA01 2140 A09 A G . PASS . GT 0|1
ChrA01 2320 A10 C T . PASS . GT 0|1
ChrB01 700 B01 T C . PASS . GT 0|1
ChrB01 880 B02 A C . PASS . GT 0|1
ChrB01 1060 B03 C G . PASS . GT 0|1
ChrB01 1240 B04 G T . PASS . GT 0|1
ChrB01 1420 B05 T A . PASS . GT 0|1
ChrB01 1600 B06 A G . PASS . GT 0|1
ChrB01 1780 B07 C T . PASS . GT 0|1
ChrB01 1960 B08 G A . PASS . GT 1|0
ChrB01 2140 B09 T C . PASS . GT 1|0
ChrB01 2320 B10 A C . PASS . GT 1|0
```

## Command

```
easyhap analyze --vcf allotetraploid_subgenomes.vcf --gff allotetraploid_subgenomes.gff3 -
-region ChrA01:1-3000 --mode copy --plot --plot-format png --outdir out_allo_A
easyhap analyze --vcf allotetraploid_subgenomes.vcf --gff allotetraploid_subgenomes.gff3 -
-region ChrB01:1-3000 --mode copy --plot --plot-format pdf,png --outdir out_allo_B
```

## Results



This example shows that EasyHap can analyze phased haplotypes independently in different allotetraploid subgenomes, reconstructing copy-specific haplotypes within each chromosome region.

## 6.4. Highly heterozygous unphased diploid: genotype mode works

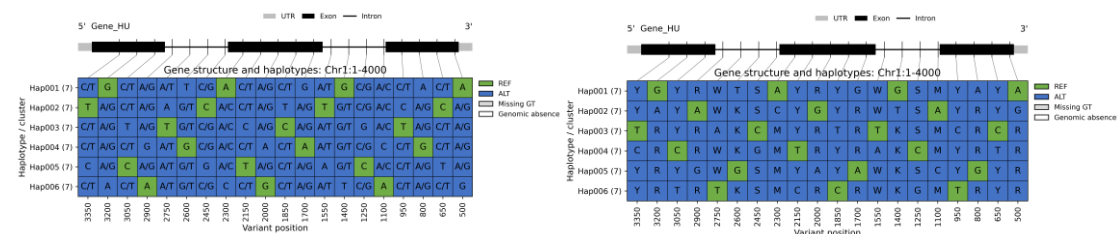
Input VCF: diploid\_high\_heterozygosity\_unphased.vcf

```
##fileformat=VCFv4.2
##contig=<ID=Chr1>
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT HUH1_01
Chr1 500 hu01 A G . PASS . GT 0/1
Chr1 650 hu02 C T . PASS . GT 1/0
Chr1 800 hu03 G A . PASS . GT 0/0
Chr1 950 hu04 T C . PASS . GT 1/0
Chr1 1100 hu05 A C . PASS . GT 1/1
Chr1 1250 hu06 C G . PASS . GT 1/0
Chr1 1400 hu07 G T . PASS . GT 0/1
Chr1 1550 hu08 T A . PASS . GT 1/0
Chr1 1700 hu09 A G . PASS . GT 0/0
Chr1 1850 hu10 C T . PASS . GT 1/0
Chr1 2000 hu11 G A . PASS . GT 1/1
Chr1 2150 hu12 T C . PASS . GT 1/0
Chr1 2300 hu13 A C . PASS . GT 0/1
Chr1 2450 hu14 C G . PASS . GT 1/0
Chr1 2600 hu15 G T . PASS . GT 0/0
Chr1 2750 hu16 T A . PASS . GT 1/0
Chr1 2900 hu17 A G . PASS . GT 1/1
Chr1 3050 hu18 C T . PASS . GT 1/0
Chr1 3200 hu19 G A . PASS . GT 0/1
Chr1 3350 hu20 T C . PASS . GT 1/0
```

### Command

```
# genotype
easyhap analyze --vcf diploid_high_heterozygosity_unphased.vcf --gff
diploid_high_heterozygosity_unphased.gff3 --region Chr1:1-4000 --mode genotype -
-plot --plot-format png --outdir out_highhet
# change genotype to iupac
easyhap analyze --vcf diploid_high_heterozygosity_unphased.vcf --gff
diploid_high_heterozygosity_unphased.gff3 --region Chr1:1-4000 --mode genotype -
hetero-policy iupac --plot --plot-format png --outdir out_highhet_iupac
```

### Results



This example shows that EasyHap can analyze highly heterozygous unphased diploid genotypes in genotype mode, with heterozygous sites represented either as genotype states or IUPAC ambiguity codes.

## 6.5. PLINK 1 binary input

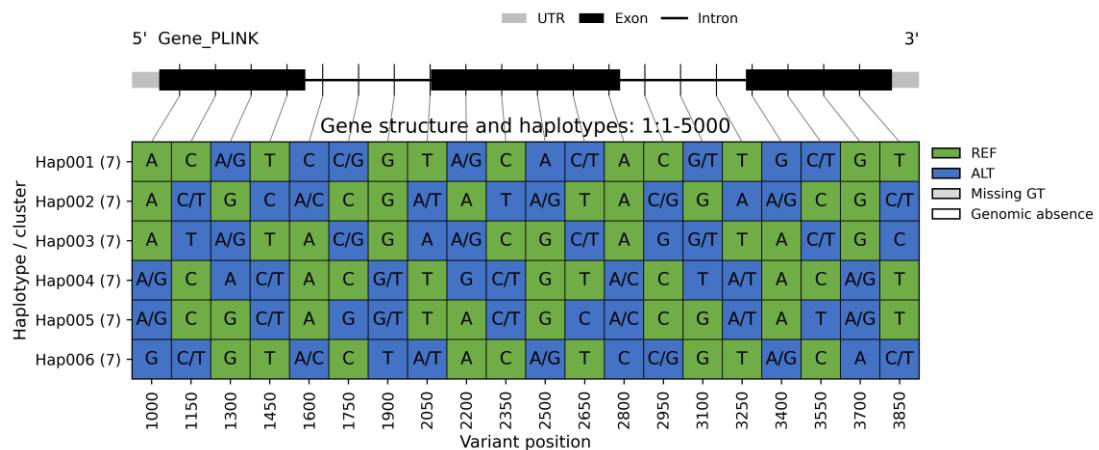
Input bfile: plink\_toy.bed/bim/fam

The example files could be retrieved from [https://github.com/Guangqi-He/EasyHap/blob/main/examples/reviewer\\_validation.zip](https://github.com/Guangqi-He/EasyHap/blob/main/examples/reviewer_validation.zip).

### Command

```
easyhap analyze --bfile plink_toy --gff plink_toy.gff3 --region 1:1-5000 --mode genotype --plot --plot-format png --outdir out_plink
```

### Results



This example shows that EasyHap can directly analyze PLINK binary genotype files (BED/BIM/FAM) in genotype mode for regional haplotype analysis and visualization.